PHYS 661 Fall 2024

Problem set 1, due 9/5 2024. Each problem is 10 pts unless otherwise stated. Submit your homework either (1) via Brightspace or by (2) emailing it directly to the TA at gbowling@purdue.edu.

1 Pauli matrices

Show that for Pauli matrices

$$e^{i\theta \mathbf{n}\cdot\boldsymbol{\sigma}} = I\cos\theta + i\mathbf{n}\cdot\boldsymbol{\sigma}\sin\theta\,,\ \text{where } \mathbf{n}^2 = 1, \tag{1}$$

and

$$e^{i\frac{1}{2}\theta\sigma_y}(\sigma_x, \sigma_y, \sigma_z)e^{-i\frac{1}{2}\theta\sigma_y} = (\sigma_x, \sigma_y, \sigma_z)\cos\theta + (\sigma_z, 0, -\sigma_x)\sin\theta + (0, \sigma_y, 0)(1-\cos\theta) \tag{2}$$

Suppose we have a Hamiltonian $H = \begin{pmatrix} a & b \\ b & -a \end{pmatrix}$ with $a, b \in \mathbb{R}$. What are the eigenvalues $E_\pm$? Find the unitary transformation U that diagonalizes H, i.e.,

$$U^\dagger H U = \begin{pmatrix} E_+ & 0 \\ 0 & E_- \end{pmatrix}. \tag{3}$$

2 1D Harmonic oscillator (Sakurai 5.1)

A simple harmonic oscillator in 1D is subjected to a perturbation

$$V = bx\,, \tag{4}$$

where $b \in \mathbb{R}$ is a constant.

1. Calculate the energy shift of the ground state to the lowest non-vanishing order.
2. Solve the problem exactly and compare your answer to 1.

3 Potential well (Sakurai 5.2)

Consider a 1D potential well with infinite walls at $x = 0, L$. The bottom is not flat, but increases linearly from 0 at $x = 0$ to V at $x = L$. Find the first-order shift in the energy levels as a function of principal quantum number n.

4 2D Harmonic oscillator (Sakurai 5.7)

Consider isotropic harmonic oscillator in 2D, with Hamiltonian

$$H = \frac{p_x^2 + p_y^2}{2m} + \frac{1}{2} m\omega^2 (x^2 + y^2)\,. \tag{5}$$

1. Calculate the energies of the 3 lowest-lying states? Is there any degeneracy?
2. Now apply a perturbation $V = \delta m\omega^2 xy$ where $\delta \ll 1$ is a dimensionless real number. Find the zeroth order energy eigenstate and the corresponding energy to first order for each of the three lowest-lying states.
3. Solve $H + V$ exactly and compare with the perturbation results obtained in 2.